Anxin (Bob) Guo

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Education

Northwestern University, Evanston, Illinois — September 2023 – Present
Ph.D. student in Computer Science. Cumulative GPA: 3.867/4.000.

Northwestern University, Evanston, Illinois — September 2019 – June 2023
B.A. in Mathematics and M.S. in Computer Science. Undergraduate cumulative GPA: 3.821/4.000.

Research Interests

Theoretical computer science and the foundations of machine learning, with interests in the capabilities and limitations of learning and computation, and how language models learn, memorize, and reason.

Research Projects and Papers

Manuscripts and Ongoing Projects

Learning Rules, Memorizing Facts: An Information-Theoretic View of Hallucination
Gengze Xu, **Anxin Guo**, Yifei Hu, Xiangyu Wu, Yong Liu, and Ziqiao Wang.
Preprint, 2026. arXiv.

How Much Do You Trust A Dog? Quantifying Hint-Following in Reasoning Language Models
Anxin Guo, Chirag Pabbaraju, Sara Ahmadian, Kamesh Munagala, and Aravindan Vijayaraghavan.
Preprint, 2026. arXiv.

Regularized Transport Weights: a Principled Approach for Active Domain Adaptation
Anxin Guo, Tanmay Sinha, Dionysis Arvanitakis, Vaidehi Srinivas, Alex Drlica-Wagner, Alexander Ji, and Aravindan Vijayaraghavan.
Manuscript.

Leanify: Verified Proof Compression and Reusable Lemma Discovery
Ruize Xu, **Anxin Guo**, Yi Gu, Tianchong Jiang, Chenxiao Yang, Tianhao Wang, and Zhiyuan Li.
Manuscript.

Copying Under Repetition: From Prefix Matching to Position Hashing
Anxin Guo, Tianlong Huang, Chenxiao Yang, and Zhiyuan Li.
Preprint, 2026. arXiv.

- Demonstrated that frontier LLMs struggle to autoregressively copy strings with extreme repetition.
- Proved impossibility results for a class of transformers spanning NoPE and a practical implementation of ALiBi: they cannot correctly terminate the copy task even with depth $N^{0.01}$, near-logarithmic precision, and arbitrary width.
- Designed a modular attention head, a plug-in augmentation using value rotation with integer period p to encode query-key distance through the value stream.
- Showed, both theoretically and empirically, that the augmented architecture can copy arbitrary strings using logarithmic width and $\log \log N$ precision.

Publications

Hallucination is a Consequence of Space-Optimality: A Rate-Distortion Theorem for Membership Testing

Anxin Guo and Jingwei Li.

International Conference on Machine Learning (ICML), 2026, spotlight, top 2.2%. arXiv.

- Defined membership testing as a unifying abstraction for approximate set membership, Bloom-filter-style error metrics, and factual errors in language models.
- Established the optimal space-error tradeoff in this framework, with an asymptotic lower bound governed by relative entropy, analogous to rate-distortion theory.
- Showed that optimally trained language models under cross-entropy loss with bounded memory may assign high confidence to unseen non-facts as a structural consequence of limited capacity.
- Extended known bounds for Bloom-type filters into a tight form, including regimes where false negatives are allowed.

Agnostic Learning of Arbitrary ReLU Activation under Gaussian Marginals

Anxin Guo and Aravindan Vijayaraghavan.

Conference on Learning Theory (COLT), 2025. arXiv; conference version; recorded talk.

- Gave the first polynomial-time constant-factor approximation agnostic learner for arbitrarily biased ReLU neurons under Gaussian marginals.
- Established a formal separation between Statistical Query (SQ) and Correlational Statistical Query (CSQ) models for this problem, proving limitations of gradient-based algorithms in this setting.
- Designed a method combining Thresholded PCA and Reweighted Gradient Descent to overcome limitations of standard gradient-based approaches.

To Store or Not to Store: A Graph-Theoretic Approach for Dataset Versioning

Anxin Guo, Jingwei Li, Pattara Sukprasert, Samir Khuller, Amol Deshpande, and Koyel Mukherjee.

IEEE International Parallel and Distributed Processing Symposium (IPDPS), 2024. arXiv; conference version.

- Formulated and analyzed approximation algorithms and heuristics for optimizing the storage-retrieval cost tradeoff in large-scale dataset versioning systems.
- Designed provably near-optimal algorithms for tree-like graphs by exploiting low-treewidth structure.
- Implemented and benchmarked proposed heuristics on experiments based on real-world GitHub repository data, improving performance by up to 1000x over prior methods.

Selected Talks

Simons Institute — Student spotlight October 7, 2026

Hallucination is a Consequence of Space-Optimality.
Workshop: *Trustworthy AI: From Hallucinations to Reliable Autonomy.* Video (my segment: first approximately 10 minutes).

SkAI Works-In-Progress — Joint presentation September 30, 2026

Intelligent Scheduling for Astronomical Surveys.
With Rachel Hur and Tanmay Sinha.

Google AI Study Group — Invited talk September 14, 2026

Why LLM Hallucinate? A Perspective from Space-Optimality.

IDEAL Annual Meeting, TTIC — Lightning talk September 10, 2026
Hallucination is a Consequence of Space-Optimality.

IDEAL Annual Meeting, UIC — Student talk June 9, 2025
Agnostic Learning of Arbitrary ReLU Neuron under Gaussian Marginals.

Northwestern University — CS Theory Seminar February 6, 2025
Agnostic Learning of Arbitrary ReLU Neuron under Gaussian Marginals.

NSF TRIPODS Workshop, TTIC — Lightning talk December 7, 2024
Agnostic Learning of Arbitrary ReLU Neuron under Gaussian Marginals.

Awards and Honors

Ph.D. Student Research Award, Computer Science Department, Northwestern University 2024 – 2025
Barris Award for Outstanding TA, CS 335: Introduction to the Theory of Computing Fall 2024
Junior Career Award in Mathematics, Mathematics Department, Northwestern University 2021 – 2022

Research and Teaching

IDEAL-funded summer research exchange, University of Chicago Summer 2026
Hosted by Chao Gao.

Research Intern / IDEAL-funded summer research exchange, TTIC Summer 2025
Toyota Technological Institute at Chicago. Hosted by Zhiyuan Li.

- Initiated the transformer copying project during the internship.
- Built experiments validating the failure mode of frontier language models and the proposed architecture.
- Presented at a reading group.

Teaching Assistant, CS 336: Design and Analysis of Algorithms Spring 2026
Teaching Assistant, CS 335: Introduction to the Theory of Computing Fall 2024

- Gave a guest lecture on busy beaver numbers in addition to regular TA duties.
- Awarded the Barris Award for Outstanding TA.

Service and Programs

Program Committee / Reviewer, Reliable ML from Unreliable Data, NeurIPS Workshop 2025
Volunteer, 65th IEEE Symposium on Foundations of Computer Science (FOCS) 2024
Ross Mathematics Program (Asia), student, junior counselor, counselor 2018 – 2020
Directed Reading Program, Linear Representations of Finite Groups Spring 2021
Read with Wenyuan Li.

Skills

Technology: C++, Python (NumPy, PyTorch, pandas, matplotlib), Mathematica, and Gurobi Optimizer.

Writing

I have written three Chinese-language articles on theoretical computer science for my Zhihu column.